

Consider the four data points

x	0	1	2	3
$f(x)$	1	2	9	28

Using the **Newton divided-difference algorithm**:

1. Construct the complete divided-difference table.
2. Write the degree-3 interpolating polynomial in **nested Newton form**.
3. Use your polynomial to evaluate $P_3(4)$.

The data points are

$$(x_0, f[x_0]) = (0, 1), \quad (x_1, f[x_1]) = (1, 2), \quad (x_2, f[x_2]) = (2, 9), \quad (x_3, f[x_3]) = (3, 28).$$

The first divided differences are

$$f[x_0, x_1] = \frac{2 - 1}{1 - 0} = 1,$$

$$f[x_1, x_2] = \frac{9 - 2}{2 - 1} = 7,$$

and

$$f[x_2, x_3] = \frac{28 - 9}{3 - 2} = 19.$$

The second divided differences are

$$f[x_0, x_1, x_2] = \frac{7 - 1}{2 - 0} = 3,$$

and

$$f[x_1, x_2, x_3] = \frac{19 - 7}{3 - 1} = 6.$$

The third divided difference is

$$f[x_0, x_1, x_2, x_3] = \frac{6 - 3}{3 - 0} = 1.$$

Thus, the complete divided-difference table is

x_i	$f[x_i]$	1st DD	2nd DD	3rd DD
0	1	1	3	1
1	2	7	6	
2	9	19		
3	28			

The Newton interpolating polynomial is

$$\begin{aligned} P_3(x) = & f[x_0] + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1) \\ & + f[x_0, x_1, x_2, x_3](x - x_0)(x - x_1)(x - x_2). \end{aligned}$$

Substituting the divided differences gives

$$P_3(x) = 1 + x + 3x(x - 1) + x(x - 1)(x - 2).$$

In nested form,

$$P_3(x) = 1 + x(1 + (x - 1)(3 + (x - 2))).$$

so

$$P_3(4) = 1 + 4(1 + (4 - 1)(3 + (4 - 2))) = 1 + 4(1 + 3(3 + 2)) = 1 + 4(1 + 15) = 65.$$

Hence,

$$P_3(4) = 65.$$

Definition. Divided differences Let $f : \mathbb{R} \rightarrow \mathbb{R}$. Let $x_0, x_1, \dots, x_n$ be $n + 1$ distinct points. Define $f[x_0, x_1, \dots, x_n]$.

$f[x_0, x_1, \dots, x_n]$ is the coefficient of x^n of the unique polynomial of degree at most n that interpolates the $n + 1$ data points: $(x_0, f(x_0)), \dots, (x_n, f(x_n))$. ($f[x_0, x_1, \dots, x_n]$ is a real number.)

Theorem. Recursive formula for divided differences Let $x_0, x_1, \dots, x_n$ be distinct points. Then

$$f[x_0, x_1, \dots, x_n] = \frac{f[x_1, x_2, \dots, x_n] - f[x_0, x_1, \dots, x_{n-1}]}{x_n - x_0}.$$

Let P_n be the unique polynomial of degree at most n that interpolates f at the $n + 1$ points:

$$x_0, x_1, \dots, x_n.$$

In Newton form,

$$P_n(x) = f[x_0] + f[x_0, x_1](x - x_0) + \dots + f[x_0, \dots, x_n] \prod_{j=0}^{n-1} (x - x_j).$$

The coefficient of x^n in P_n is

$$f[x_0, \dots, x_n].$$

Now let Q_{n-1} be the interpolation polynomial of degree at most $n - 1$ through the n data points:

$$(x_0, f(x_0)), \dots, (x_{n-1}, f(x_{n-1})),$$

The coefficient of x^{n-1} in Q_{n-1} is

$$f[x_0, \dots, x_{n-1}].$$

And let R_{n-1} be the interpolation polynomial of degree at most $n - 1$ through the n data points:

$$(x_1, f(x_1)), \dots, (x_n, f(x_n)).$$

The coefficient of x^{n-1} in R_{n-1} is

$$f[x_1, \dots, x_n].$$

Consider the polynomial

$$S_n(x) = \frac{(x - x_0)R_{n-1}(x) - (x - x_n)Q_{n-1}(x)}{x_n - x_0}.$$

We claim that $S_n = P_n$.

First the degree of S_n is at most n .

Second, we claim that S_n interpolates f at the $n + 1$ points:

$$x_0, x_1, \dots, x_n.$$

Indeed, for $i = 1, \dots, n - 1$, we have

$$\begin{aligned} S_n(x_i) &= \frac{(x_i - x_0)R_{n-1}(x_i) - (x_i - x_n)Q_{n-1}(x_i)}{x_n - x_0} \\ &= \frac{(x_i - x_0)f(x_i) - (x_i - x_n)f(x_i)}{x_n - x_0} \\ &= f(x_i). \end{aligned}$$

At the two endpoints,

$$S_n(x_0) = Q_{n-1}(x_0) = f(x_0),$$

and

$$S_n(x_n) = R_{n-1}(x_n) = f(x_n).$$

Thus S_n interpolates f at all $n + 1$ points $x_0, \dots, x_n$.

S_n and P_n are two polynomials of degree at most n that interpolates f at the $n + 1$ points $x_0, \dots, x_n$, thus by uniqueness of the interpolating polynomial, we have

$$S_n = P_n.$$

It remains to compare the coefficients of x^n . The coefficient of x^n in

$$(x - x_0)R_{n-1}(x)$$

is $f[x_1, \dots, x_n]$, while the coefficient of x^n in

$$(x - x_n)Q_{n-1}(x)$$

is $f[x_0, \dots, x_{n-1}]$. Therefore the coefficient of x^n in S_n is

$$\frac{f[x_1, \dots, x_n] - f[x_0, \dots, x_{n-1}]}{x_n - x_0}.$$

But $S_n = P_n$, and the coefficient of x^n in P_n is $f[x_0, \dots, x_n]$. Consequently,

$$f[x_0, \dots, x_n] = \frac{f[x_1, \dots, x_n] - f[x_0, \dots, x_{n-1}]}{x_n - x_0}.$$

Remainder theorem for polynomial interpolation. Let $x_0, x_1, \dots, x_n$ be distinct points, and let P_n be the polynomial of degree at most n that interpolates f at these points:

$$P_n(x_i) = f(x_i), \quad i = 0, 1, \dots, n.$$

Let x be any point in the domain of f . Assume that $f \in C^{n+1}(J)$, where J is an interval containing

$$x_0, x_1, \dots, x_n, x.$$

Then there exists a point $c \in J$, lying between the smallest and largest of

$$x_0, x_1, \dots, x_n, x,$$

such that

$$f(x) - P_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!} \prod_{i=0}^n (x - x_i).$$

If $x = x_i$ for some i , then

$$f(x) - P_n(x) = 0$$

and

$$\prod_{i=0}^n (x - x_i) = 0,$$

so the result is immediate. Assume, therefore, that $x \notin \{x_0, \dots, x_n\}$.

Now regard x as one additional interpolation point. By the Newton form of the interpolating polynomial through

$$x_0, x_1, \dots, x_n, x,$$

we have

$$f(x) = P_n(x) + f[x_0, x_1, \dots, x_n, x] \prod_{i=0}^n (x - x_i).$$

Consequently,

$$f(x) - P_n(x) = f[x_0, x_1, \dots, x_n, x] \prod_{i=0}^n (x - x_i).$$

It remains to identify the divided difference $f[x_0, x_1, \dots, x_n, x]$.

Define

$$h(t) = f(t) - P_n(t) - f[x_0, x_1, \dots, x_n, x] \prod_{i=0}^n (t - x_i).$$

For each $i = 0, 1, \dots, n$,

$$h(x_i) = f(x_i) - P_n(x_i) - f[x_0, \dots, x_n, x] \prod_{j=0}^n (x_i - x_j) = 0,$$

because

$$P_n(x_i) = f(x_i)$$

and the product contains the factor $x_i - x_i$.

Moreover, from the preceding Newton interpolation formula,

$$h(x) = 0.$$

Thus h has $n + 2$ distinct zeros:

$$x_0, x_1, \dots, x_n, x.$$

By repeated applications of Rolle's theorem, there exists a point c , lying between the smallest and largest of these points, such that

$$h^{(n+1)}(c) = 0.$$

Since P_n has degree at most n ,

$$P_n^{(n+1)}(t) = 0.$$

Also,

$$\omega(t) = \prod_{i=0}^n (t - x_i)$$

is a monic polynomial of degree $n + 1$. Therefore,

$$\omega^{(n+1)}(t) = (n + 1)!.$$

Hence

$$h^{(n+1)}(t) = f^{(n+1)}(t) - (n + 1)! f[x_0, x_1, \dots, x_n, x].$$

Evaluating at $t = c$ gives

$$0 = f^{(n+1)}(c) - (n + 1)! f[x_0, x_1, \dots, x_n, x].$$

Thus

$$f[x_0, x_1, \dots, x_n, x] = \frac{f^{(n+1)}(c)}{(n + 1)!}.$$

Substituting this into the error expression obtained above yields

$$\boxed{f(x) - P_n(x) = \frac{f^{(n+1)}(c)}{(n + 1)!} \prod_{i=0}^n (x - x_i).}$$